

Forms and React Events

Today You Will Learn

Event Handling

Forms in React

Controlled Forms

Multiple Inputs

Form Validation

Mini Projects

📌 **Learning Outcome:** By the end of today's session, you will be able to build interactive forms, capture user input, validate data, and handle events in React applications.

Event Handling

What is Event Handling?

An event is an action performed by the user.

Examples

- Clicking a button
- Typing in an input
- Submitting a form
- Hovering over an element

Common React Events

onClick onChange onSubmit onMouseOver onKeyDown onFocus onBlur

Real World

- Login Button
- Delete Button
- Add to Cart
- Search Button

Example

```
function App() {  
  function handleClick() {  
    alert("Button Clicked");  
  }  
  return (  
    <button onClick={handleClick}>Click Me</button>  
  );  
}
```

Forms in React

Simple Form

```
function App() {  
  return (  
    <form>  
      <input type="text" />  
      <button type="submit">Submit</button>  
    </form>  
  );  
}
```

⚠ **Problem:** React doesn't know the input value. We need **State**.

Controlled Forms

What is a Controlled Component?

The input value is controlled by React state.

```
import { useState } from "react";

function App() {
  const [name, setName] = useState("");
  return (
    <>
      <input
        type="text"
        value={name}
        onChange={(e) => setName(e.target.value)}
      />
      <h1>Hello {name}</h1>
    </>
  );
}
```

Benefits

Easy Validation

Real-time Updates

Better User Experience

Multiple Inputs

Instead of multiple states, store everything in one object.

```
const [formData, setFormData] = useState({  
  name: "",  
  email: "",  
  password: ""  
});
```

Update Input

```
const handleChange = (e) => {  
  setFormData({  
    ...formData,  
    [e.target.name]: e.target.value  
  });  
};
```

Benefits

Cleaner Code

Easy Maintenance

Best Practice

Form Validation

Validate Before Submit

```
const handleSubmit = (e) => {  
  e.preventDefault();  
  if (formData.name === "") {  
    alert("Name Required");  
    return;  
  }  
  alert("Form Submitted");  
};
```

Common Validations

- Required Fields
- Email Format
- Password Length
- Phone Number
- Confirm Password

Best Practice

Always validate on both:

- Frontend
- Backend

Form Events

onSubmit — Prevent Page Reload

```
e.preventDefault();
```

onChange

onFocus

```
<input onFocus={() => console.log("Focused")} />
```

onBlur

```
<input onBlur={() => console.log("Left Input")} />
```

Real World

Login Forms

Search Forms

Registration Forms

Payment Forms



Join thousands of innovators building the future.

Create your account and unlock exclusive features, resources, and opportunities.

Create your account

Fill in your details to get started.

Full Name

 Alexandra Morgan

Email Address

 alexandra.morgan@example.com

Phone Number

 +1 (555) 123-4567

Password

 

Must be at least 8 characters with a mix of letters, numbers & symbols.

Confirm Password

 

Date of Birth

 April 12, 1990

Country

 United States 

☒ I agree to the [Terms of Service](#) and [Privacy Policy](#).

Create Account

OR

 Sign up with Google

Already have an account? [Sign in](#)

Registration Form (Mini Project)

Build Registration Form

Fields

- Full Name
- Email
- Mobile Number
- Password
- Confirm Password

Features

- Controlled Components
- Form Validation
- Bootstrap Form
- Submit Button
- Show entered data below the form

Concepts Used



useState



Event Handling



Multiple Inputs



Validation

Login Form (Mini Project)

Build Login Form

Fields

- Email
- Password

Features

- Controlled Form
- Login Button
- Show Welcome Message after Submit
- Validate Empty Fields

Example

```
if (email === "") {  
  alert("Enter Email");  
}
```

Bootstrap Components

Form

Input

Button

Card

Homework Assignment

Homework — Contact Form

Create a responsive Contact Form using Bootstrap.

Fields

- Full Name
- Email
- Phone Number
- Subject
- Message

Validation Rules

- Name Required
- Valid Email
- Mobile should contain 10 digits
- Subject Required
- Message should contain at least 20 characters

Bonus Challenge

After successful submission: Display a Bootstrap Alert.

- ✓ Thank You! Your message has been sent successfully.

Features

- ✓ Controlled Components
- ✓ Multiple Inputs
- ✓ Form Validation
- ✓ Bootstrap Styling
- ✓ Responsive Design

Challenge 2

Clear all form fields after successful submission.

Hint

```
setFormData({
  name: "",
  email: "",
  phone: "",
  subject: "",
  message: ""
});
```

Submission Checklist

- ✓ Event Handling
- ✓ Controlled Components
- ✓ Multiple Inputs
- ✓ Form Validation
- ✓ Bootstrap Form
- ✓ Responsive Design
- ✓ Clear Form After Submit



Expected Outcome: Students will learn how to handle user events, manage form data using React state, validate inputs, and build responsive registration, login, and contact forms. These concepts are fundamental for authentication systems, user registration, profile management, and nearly every production React application.